Name: Date:

Standards: MA.5.GR.1.1; MA.5.GR.4.1-4.2

Show your work in the box for each problem.

Write your final answer on the answer line (or in the blanks shown).

Use the words and strategies from Unit 11: geometry.

1 Recall: Angle Types

List the four types of angles and their degree ranges.

Name each angle type and give its degree range.

Acute: 0°–90° (not including 90°); Right: exactly 90°; Obtuse: 90°–180° (not including those endpoints); Straight: exactly 180°

2 Recall: Quadrilateral Properties

Name a quadrilateral that has exactly one pair of parallel sides.

Name the shape and give one other property.

Trapezoid. One other property: the two angles on the same leg are supplementary (sum to 180°).

3 Application: Find the Angle

A triangle has angles of 47° and 81°.

Find the missing angle and classify the triangle by its angles.

Missing angle = 180° – 47° – 81° = 52°. All angles are less than 90° — acute triangle.

4 Application: Quadrilateral Angles

A quadrilateral has angles 105°, 88°, 72°, and an unknown angle.

Find the unknown angle.

360° – 105° – 88° – 72° = 95°.

5 Application: Classify Two Triangles

Triangle 1: sides 5 cm, 5 cm, 8 cm; angles 68°, 68°, 44°. Triangle 2: sides 6 cm, 8 cm, 10 cm; angles 37°, 53°, 90°.

Classify each triangle by sides and by angles.

Triangle 1: Isosceles, Acute. Triangle 2: Scalene, Right.

6 Word Problem: Flag Design

A triangular flag has a right angle and one angle of 32°.

Find all three angles. What type of triangle is it by sides if the two legs are different lengths?

Angles: 90°, 32°, 58°. By angles: Right. By sides: Scalene.

7 Word Problem: Park Path

A park path forms a quadrilateral with angles 90°, 85°, 95°, and an unknown angle.

Find the unknown angle. Is the path shaped like a rectangle? Explain.

Unknown = 360° – 90° – 85° – 95° = 90°. No — a rectangle requires all four angles to be 90°, but two angles are 85° and 95°

8 Find the Mistake

A student says an isosceles triangle has angles 50°, 50°, and 90°.

Find the error.

50° + 50° + 90° = 190° ≠ 180°. This is not a valid triangle. A valid right isosceles triangle has angles 90°, 45°, and 45°.

9 Multi-Step: Composite Shape

A shape is made of a rectangle with a triangle on top. The rectangle has four right angles. The triangle has two angles of 65° each.

Find the third angle of the triangle. What is the total number of angles in the composite shape?

Triangle's third angle = 180° – 65° – 65° = 50°. Total angles: rectangle (4) + triangle (3) = 7 angles.

10 Reasoning: Sophia's Shapes

Sophia says every rhombus is a parallelogram, but not every parallelogram is a rhombus.

Is she correct? Use properties to justify.

Yes. A parallelogram has opposite sides parallel and equal. A rhombus also has all four sides equal, so it satisfies the paralle